

WARNING: MISBEHAVIOR AT EXAM TIME WILL LEAD TO SERIOUS CONSEQUENCE.

SCUT Final Exam

2023-2024-2 《Calculus II》 Exam Paper B

- Notice: 1. Make sure that you have filled the form on the left side of seal line.
2. Write your answers on the exam paper.
3. This is a close-book exam.
4. The exam with full score of 100 points lasts 120 minutes.

Question No.	1	2	3	Sum
Score				

1. Answer the following questions ($4 \times 7 = 28$ points):

- (1) Find symmetric equations for the tangent line to the intersection curve determined by two

cylindrical surfaces $\begin{cases} x^2 + y^2 = 1 \\ x^2 + z^2 = 1 \end{cases}$ at point $P\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

Solu.

$$\begin{cases} 2x + 2y \cdot y' = 0 \\ 2x + 2z \cdot z' = 0 \end{cases}$$

$$y'(P) = z'(P) = -1,$$

$$\vec{v} = \langle 1, -1, -1 \rangle,$$

$$\text{symmetric equations: } \frac{x-1/\sqrt{2}}{1} = \frac{y-1/\sqrt{2}}{-1} = \frac{z-1/\sqrt{2}}{-1}.$$

- (2) Find the directional derivative of $f(x, y, z) = \ln(x + \sqrt{y^2 + z^2})$ at $P(1, 0, 1)$ in the direction towards $Q(3, -2, 2)$.

Solu.

$$f_x(P) = \frac{1}{2}, f_y(P) = 0, f_z(P) = \frac{1}{2}, \vec{u} = \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \langle \frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \rangle$$

$$D_u f(P) = \langle \frac{1}{2}, 0, \frac{1}{2} \rangle \cdot \langle \frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \rangle = \frac{1}{2}.$$

(3) Calculate line integral $\int_L (x+y)^2 ds$, where L is the curve satisfying $L: x^2 + y^2 = 1$.

Solu.

$$\begin{aligned}\int_L (x+y)^2 ds &= \int_L (x^2 + y^2 + 2xy) ds = \int_L (x^2 + y^2) ds + 0 \\ &= \int_L 1 ds = 2\pi.\end{aligned}$$

(4) Find the minimum and maximum values of the function $f(x, y) = x^2 + y^2$ on the closed bounded set $S = \{(x, y) | (x-1)^2 + (y-1)^2 \leq 4\}$.

Solu.

(1) In $\text{int } S = \{(x, y) | (x-1)^2 + (y-1)^2 < 4\}$, singular point $(0, 0)$, $f(0, 0) = 0$;

(2) On the set $\partial S = \{(x, y) | (x-1)^2 + (y-1)^2 = 4\}$,

$$L(x, y, \lambda) = x^2 + y^2 + \lambda[(x-1)^2 + (y-1)^2 - 4],$$

$$L_x = L_y = L_\lambda = 0, \text{ singular points } (1 + \sqrt{2}, 1 + \sqrt{2}), (1 + \sqrt{2}, 1 - \sqrt{2}),$$

$$f(1 + \sqrt{2}, 1 + \sqrt{2}) = 6 + 4\sqrt{2}, \quad f(1 - \sqrt{2}, 1 - \sqrt{2}) = 6 - 4\sqrt{2};$$

$$\text{maximum value } f(1 + \sqrt{2}, 1 + \sqrt{2}) = 6 + 4\sqrt{2},$$

$$\text{minimum value } f(0, 0) = 0.$$

2. Answer the following questions (8×6=48 points):

(1) Find double integral $\iint_D y^2 dA$, where D is a closed region enclosed by a straight line $y = x$

and a parabola $x = y^2 - 2$.

Solu.

$$\begin{aligned}I &= \int_{-1}^2 \left[\int_{y^2-2}^y y^2 dx \right] dy \\ &= \int_{-1}^2 y^2(y - y^2 + 2) dy = \frac{63}{20}.\end{aligned}$$

(2) Find the iterated integral $I = \int_0^1 \int_x^1 x^2 e^{-y^2} dy dx$.

$$\text{Solu. } I = \int_0^1 \int_0^y x^2 e^{-y^2} dx dy = \int_0^1 e^{-y^2} \cdot \frac{y^3}{3} dy$$

$$= \frac{e-2}{6e}.$$

(3) Evaluate $\iiint_{\Omega} \sqrt{x^2 + y^2} dV$, where Ω is bounded by $y + z = 4$, $z = 0$ and $x^2 + y^2 = 16$.

$$\begin{aligned} \text{Solu. } \iiint_{\Omega} \sqrt{x^2 + y^2} dV &= \int_0^{2\pi} \left[\int_0^4 \left[\int_0^{4-r\sin\theta} r dz \right] r dr \right] d\theta \\ &= \frac{512}{3} \pi. \end{aligned}$$

(4) Calculate $I = \int_L (e^x \sin y - y^3) dx + (e^x \cos y + x^3) dy$, where L is the half circle from point $A(0, -a)$ to point $B(0, a)$ along the curve $x = -\sqrt{a^2 - y^2}$ ($a > 0$).

Solu.

$$\begin{aligned} I &= \oint_{L+\overline{BA}} - \int_{\overline{BA}} \\ &= - \iint_D [(e^x \cos y + x^3)_x - (e^x \sin y - y^3)_y] dx dy - \int_{\overline{BA}} (e^x \cos y + x^3) dy \\ &= - \iint_D (3x^2 + 3y^2) dx dy - \int_a^{-a} e^0 \cdot \cos y dy \\ &= -\frac{3\pi}{4} a^4 + 2 \sin a. \end{aligned}$$

(5) Calculate surface integral $\iint_{\Sigma} z dS$, where Σ is the finite part of a plane $x + y + z = 4$ that is cut off by a cylindrical surface $x^2 + y^2 = 1$.

Solu.

$$\begin{aligned} \iint_{\Sigma} z dS &= \iint_{x^2+y^2 \leq 1} (4 - x - y) \sqrt{1 + (-1)^2 + (-1)^2} dx dy \\ &= 4\sqrt{3}\pi. \end{aligned}$$

(6) Classify the following series as absolutely convergent, conditionally convergent or divergent: $\sum_{n=3}^{\infty} (-1)^n \frac{n}{n^2 - n - 2}$.

Solu. $a_n = \frac{n}{n^2 - n - 2}$, $a_n > a_{n+1}$, $a_n \rightarrow 0$ ($n \rightarrow \infty$), by AST, $\sum_{n=3}^{\infty} (-1)^n \frac{n}{n^2 - n - 2}$ converges.

$$\sum_{n=3}^{\infty} \left| (-1)^n \frac{n}{n^2 - n - 2} \right| = \sum_{n=3}^{\infty} \frac{n}{n^2 - n - 2},$$

$$\because \frac{n}{n^2 - n - 2} > \frac{n}{2n^2} = \frac{1}{2n}, \sum_{n=3}^{\infty} \frac{1}{2n} \text{ diverges,}$$

$$\therefore \sum_{n=3}^{\infty} \left| (-1)^n \frac{n}{n^2 - n - 2} \right| \text{ diverges,}$$

Therefore, $\sum_{n=3}^{\infty} (-1)^n \frac{n}{n^2 - n - 2}$ is conditionally convergent.

(7) Expand the function $f(x) = \frac{1}{x^2 - 3x - 4}$ into Maclaurin series (the power series of x).

Solu.

$$\begin{aligned} f(x) &= -\frac{1}{5} \left(\frac{1}{1 - (-x)} + \frac{1}{4} \cdot \frac{1}{1 - \frac{x}{4}} \right) \\ &= -\frac{1}{5} \left[\sum_{n=0}^{\infty} (-x)^n + \frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{x}{4} \right)^n \right] \quad (|x/4| < 1, |-x| < 1) \\ &= -\frac{1}{5} \left[\sum_{n=0}^{\infty} \left((-1)^n + \frac{1}{4^{n+1}} \right) x^n \right], \quad (-1 < x < 1). \end{aligned}$$

(8) Find the convergent radius, convergent set and sum function $S(x)$ for the series

$$\sum_{n=2}^{\infty} n(n-1)x^n, \text{ and then find the sum of the series } \sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}.$$

Solu. $R = 1$,

convergent set $(-1, 1)$

$$S(x) = \sum_{n=2}^{\infty} n(n-1)x^n = x^2 \sum_{n=2}^{\infty} (x^n)'' = x^2 \left(\sum_{n=2}^{\infty} x^n \right)'' = x^2 \left(\frac{x^2}{1-x} \right)'' = \frac{2x^2}{(1-x)^3}, x \in (-1, 1).$$

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n} = S\left(\frac{1}{2}\right) = 4.$$

3. Answer the following questions (4×6=24 points):

- (1) If $\varphi(u, v)$ is differentiable, and a, b, c are **non-zero** constants, the function $z = z(x, y)$ is determined by the equation $\varphi(cx - az, cy - bz) = 0$, find $a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y}$.

Solu. $G(x, y, z) = \varphi(cx - az, cy - bz)$,

$$\begin{aligned} G_x &= c\varphi'_1, & G_y &= c\varphi'_2, & G_z &= -a\varphi'_1 - b\varphi'_2, \\ \frac{\partial z}{\partial x} &= -\frac{G_x}{G_z} = \frac{c\varphi'_1}{a\varphi'_1 + b\varphi'_2}, & \frac{\partial z}{\partial y} &= -\frac{G_y}{G_z} = \frac{c\varphi'_2}{a\varphi'_1 + b\varphi'_2}, \\ a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} &= a \cdot \frac{c\varphi'_1}{a\varphi'_1 + b\varphi'_2} + b \cdot \frac{c\varphi'_2}{a\varphi'_1 + b\varphi'_2} = c. \end{aligned}$$

- (2) If $f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases}$ then please

(1) find $f_x(x, y)$, $f_y(x, y)$ and determine whether they are continuous or not at point $(0, 0)$;

(2) determine $f(x, y)$ is differentiable or not at point $(0, 0)$.

Solu.

$$\begin{aligned} f_x(x, y) &= \begin{cases} 2x \sin \frac{1}{x^2 + y^2} - \frac{2x}{x^2 + y^2} \cos \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases} \\ f_y(x, y) &= \begin{cases} 2y \sin \frac{1}{x^2 + y^2} - \frac{2y}{x^2 + y^2} \cos \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases} \\ \frac{\Delta f - [f_x(0, 0)\Delta x + f_y(0, 0)\Delta y]}{\sqrt{\Delta x^2 + \Delta y^2}} &= \sqrt{\Delta x^2 + \Delta y^2} \sin \frac{1}{\sqrt{\Delta x^2 + \Delta y^2}} \rightarrow 0 (\sqrt{\Delta x^2 + \Delta y^2} \rightarrow 0), \end{aligned}$$

$f(x, y)$ is differentiable at point $(0, 0)$.

$f_x(x, y)$, $f_y(x, y)$ are not continuous at point $(0, 0)$.

- (3) Find the general solution to the differential equation $xdy = (y - x)dx$.

Solu.

$$y' - \frac{1}{x}y = -1,$$

$$\begin{aligned} y &= e^{-\int(-\frac{1}{x})dx} \left[C + \int -e^{\int(-\frac{1}{x})dx} dx \right] \\ &= x(C - \ln|x|). \end{aligned}$$

(4) Find the general solution to the differential equation $y'' - y = 2e^x - x^2$.

Solu.

$$r = \pm 1,$$

$$Y = C_1 e^x + C_2 e^{-x}$$

$$y^* = axe^x + (bx^2 + cx + d), a = 1, b = 1, c = 0, d = 2$$

$$y^* = xe^x + x^2 + 2$$

$$\text{general solution: } y = C_1 e^x + C_2 e^{-x} + xe^x + x^2 + 2.$$